Exercise 6

Date Functions (Advanced SQL)

Table 1: Employees

emp_id	name	hire_date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1: Add 6 months to each employee's hire date using DATEADD().

```
SELECT emp_id, name, hire_date, DATEADD(MONTH, 6, hire_date) AS hire_plus_6_months
FROM Employees;
```

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

student_id	name	dob
101	Maya	2005-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

Q2: Use DATEDIFF() to find age in days from dob to today

```
SELECT student_id, name, dob, DATEDIFF(DAY, dob, '2026-04-25') AS age_in_days
FROM Students;
```

student_id	name	age_in_days
101	Maya	7563
102	Ethan	7816
103	Sienna	7346

Table 3 : Events

event_id	event_name	event_date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

Q3: Find how many days are left until each event using DATEDIFF().

SELECT event_id, event_name, DATEDIFF(DAY, '2026-04-25', event_date) AS days_left
FROM Events;

event_id	event_name	days_remaining
1	Seminar	-679
2	Workshop	-236
3	Hackathon	-460

Table 4 : Invoices

invoice_id	issue_date	due_date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Q4: Calculate the number of days between issue_date and due_date.

SELECT invoice_id, issue_date, due_date, DATEDIFF(DAY, issue_date, due_date) AS days_between
FROM Invoices;

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5 : Courses

course_id	name	start_date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

Q5: Format start_date as 'Month YYYY' using TO_CHAR().

```
SELECT course_id, name, TO_CHAR(start_date, 'Month YYYY') AS formatted_date
FROM courses;
```

course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Membership

member_id	start_year	start_month
1	2023	5
2	2022	11

Q6: Create full date from parts using DATE_FROM_PARTS().

```
SELECT member_id,
        DATEFROMPARTS(start_year, start_month, start_day) AS full_date
FROM Membership;
```

member_id	full_date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

sub_id	plan	renewal_date
11	Basic	2025-01-01
12	Premium	2025-03-15

Q7: Extend each renewal_date by 1 year using DATEADD().

```
SELECT sub_id, plan, DATEADD(Year, 1, renewal_date) AS extended_renewal_date
FROM subscriptions;
```

sub_id	plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8: Orders

order_id	order_date
1001	2025-04-15
1002	2025-04-15

Q8: Show current date and difference from order_date. Use CURRENT_DATE and DATEDIFF().

```
SELECT order_id, order, date,
        CURRENT_DATE AS today_date,
        DATEDIFF(DAY, order_date, CURRENT_DATE) AS days_since_order
FROM Orders;
```

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-25	375
1002	2025-04-10	2026-04-25	380

Table 9: Trainings

training_id	topic	training_date
1	Safety	2025-01-10
2	Compliance	2025-02-20

Q9: Extract the year from training_date using DATEPART() or EXTRACT().

```
SELECT training id, topic, EXTRACT(YEAR FROM training_date) AS training_year
FROM Trainings;
```

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Table 10: Blogs_Posts

post_id	title	published_on
1	SQL Tips	2025-04-01 10:15:00
2	Data Cleaning	2025-04-12 16:45:00

Q10: Extract hour and minute from published_on.

```
SELECT post_id,
    title,
    EXTRACT (HOUR FROM published_on) AS hour_published,
    EXTRACT (MINUTE FROM published_on) AS minute_published
FROM ~~Blogs_Posts~~ Blog_Posts;
```

post_id	title	hour_published	minute_published
1	SQL Tips	10	15
2	Data Cleaning	16	45

Table 11: Drivers

driver_id	license_expiry
301	2025-08-10
302	2023-12-31

Q11: Calculate days left until license using DATEDIFF() and today's date

```
SELECT driver_id, license_expiry, DATEDIFF(DAY, '2026-04-25', license_expiry) AS days_left
FROM Drivers;
```

driver_id	license_expiry	days_left
301	2025-08-10	258
302	2023-12-31	-846

Table 12: Messages

message_id	sent_timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

Q12: Display the current timestamp and calculate seconds since the message was sent.

```
SELECT message_id,
        sent_timestamp,
        CURRENT_TIMESTAMP AS current_timestamp
        DATEDIFF(SECOND, sent_timestamp, CURRENT TIMESTAMP) AS seconds_since_sent
FROM Messages;
```

message_id	sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	2026-04-25 00:00:00	32113635
2	2025-04-18 23:59:59	2026-0425 00:00:00	32140801

Table 13: Returns

return_id	return_date
901	2025-04-05
902	2025-04-01

Q13: Add 15 days to return_date using DATEADD() to show restock_date.

```
SELECT return_id, return_date, DATEADD(DAY, 15, return_date) AS restock_date
FROM Returns;
```

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14 : Assignments

assign_id	assigned_on
1	2025-03-01
2	2025-03-05

Q 14: Convert assigned_on to date using TO_DATE() (if it's stored as string)

SELECT assign_id, TO_DATE (assigned_on, 'YYYY-MM-DD') As assigned_on_date
FRom Assignments;

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Table 15 : Meetings

meeting_id	scheduled_time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

Q15: convert scheduled_time to formatted string like 'April, 19, 2025 at 2:00 PM' using TO_CHAR().

SELECT meeting_id,
TO_CHAR (scheduled time, 'Month DD, YY "at" HH12 : MI AM') As formatted meeting_tim
FRom Meetings;

meeting_id	formatted_meeting_time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM